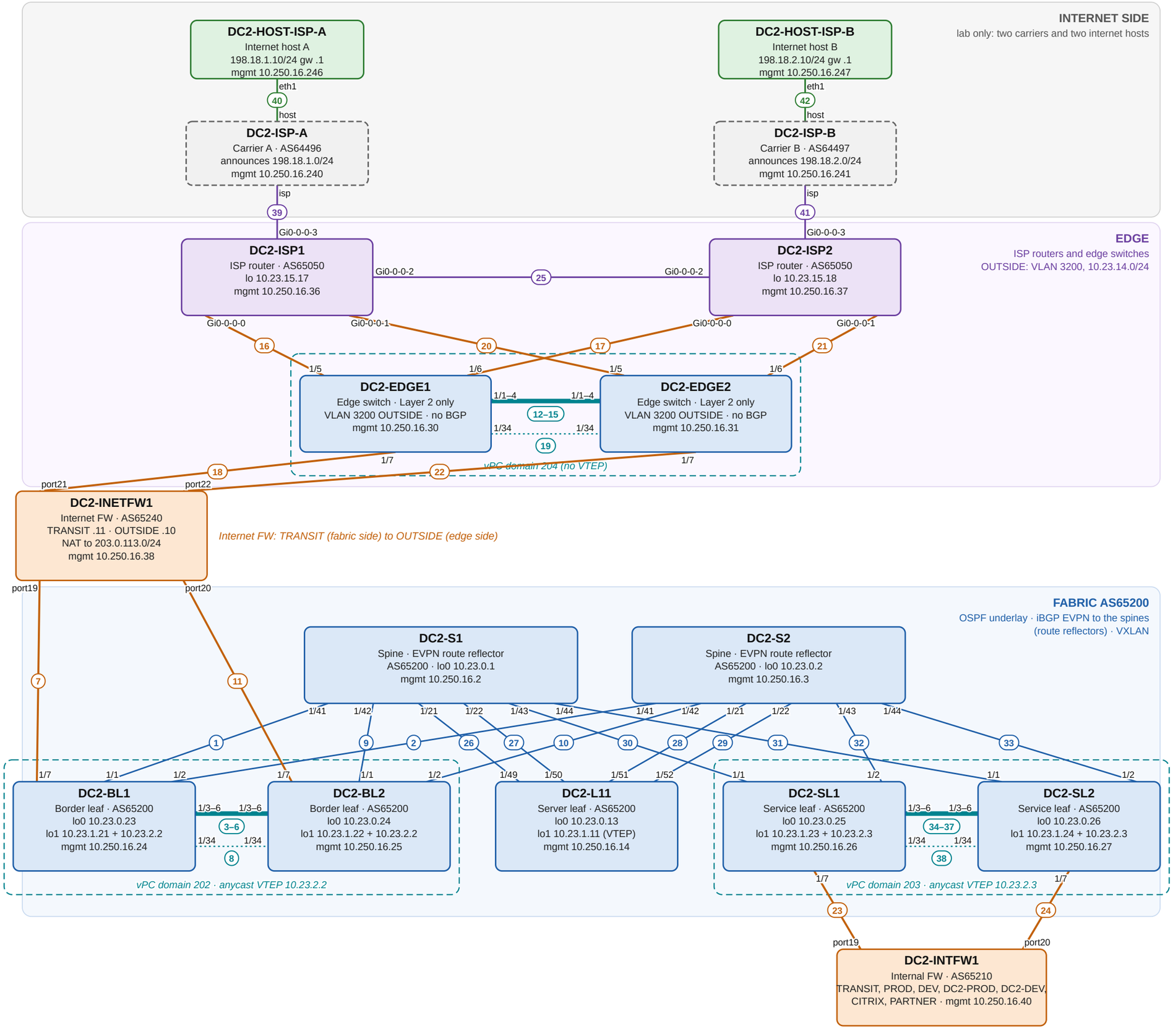
DC2 lab path: every device and cable



Legend

Boxes: what runs in the lab

- Nexus switch: N9Kv 10.6(4)M lite (vrnetlab), 8 GB, 3 vCPU
- FortiGate firewall: FRR 10.6.0 stand-in
- Cisco 8201 ISP router (IOS XR): FRR 10.6.0 stand-in
- Carrier (exists only in the lab): FRR 10.6.0
- Test host: Alpine 3.22

Lines: what each cable is

- Fabric link: routed /31, OSPF UNDERLAY area 0, MTU 9216
- vPC peer-link: port-channel1000, 4 members (drawn as one)
- vPC peer-keepalive: Eth1/34, VRF VPC-KEEPALIVE /31
- LACP bundle, one member per vPC switch: 802.1Q trunk
- Routed link outside the fabric (ISP cross-link, carriers)
- Host link
- vPC pair (domain number, anycast VTEP)
- 26 — Link number: addressing on page 2
- 1/41 — Port at that end of the cable

How to read the ports

Nexus: 1/41 = Ethernet1/41 in the config;
containerlab calls the same port eth41.
FortiGate stand-ins: port19…port22 (FortiOS names).
ISP stand-ins: Gi0-0-0-0 = Gi0/0/0/0 (IOS XR name with / replaced by -, as Linux needs).

The lab at a glance

Lab name vamd-dc2-p3 · 17 nodes · 42 links
Containers: clab-vamd-dc2-p3-<node name>
Mgmt network vamd-dc2-mgmt 10.250.16.0/24
(every box shows its mgmt address)
9 Nexus × 8 GB = 72 GB; whole lab about 80 GB.
Rendered and validated; never run.
No PROD test host: DC2-L11 has no host port.
Only DC2-L11 of the 20 server leaves is in the lab; only one unit of each firewall/ISP pair (INTFW1, INETFW1) except the ISP routers (both).
Spine configs still list all 26 EVPN clients: the sessions to absent devices stay down (expected).

Not in this lab (production DC2 has them)

L1–L10, L12–L20, BGW1/2, DCI1/2, DMZ1/2, INTFW2, INETFW2, SDWANFW1/2, VPNFW1/2, CLOUD1/2. Their SVIs exist on BL/SL with no peer.

Source of every value

lab-topology.DC2.yml and the DC2 lab configs rendered by Infrahub, main a7bcc19.
Layout drawn by hand; values copied, not typed.

DC2 lab path: addressing of every cable and device

The 42 cables (numbers match page 1; order as in lab-topology.DC2.yml)

#	End A (port)	End B (port)	Type	Addressing / VLANs / purpose
1	DC2-BL1 **1/1**	DC2-S1 **1/41**	fabric /31	10.23.4.80/31 · S1 .80, BL1 .81
2	DC2-BL1 **1/2**	DC2-S2 **1/41**	fabric /31	10.23.4.172/31 · S2 .172, BL1 .173
3	DC2-BL1 **1/3**	DC2-BL2 **1/3**	vPC peer-link	vPC 202 peer-link Po1000 · trunk 2531,2599,3001-3004,3300,3900
4	DC2-BL1 **1/4**	DC2-BL2 **1/4**	vPC peer-link	vPC 202 peer-link Po1000
5	DC2-BL1 **1/5**	DC2-BL2 **1/5**	vPC peer-link	vPC 202 peer-link Po1000
6	DC2-BL1 **1/6**	DC2-BL2 **1/6**	vPC peer-link	vPC 202 peer-link Po1000
7	DC2-BL1 **1/7**	DC2-INETFW1 **port19**	LACP bundle member	Po11 / vPC 11 ↔ agg-inside · VLAN 3001 TRANSIT 10.23.16.8/29 · BL1 .9, BL2 .10, FW .11
8	DC2-BL1 **1/34**	DC2-BL2 **1/34**	vPC keepalive	vPC 202 keepalive · VRF VPC-KEEPALIVE 10.23.17.2/31 · BL1 .2, BL2 .3
9	DC2-BL2 **1/1**	DC2-S1 **1/42**	fabric /31	10.23.4.82/31 · S1 .82, BL2 .83
10	DC2-BL2 **1/2**	DC2-S2 **1/42**	fabric /31	10.23.4.174/31 · S2 .174, BL2 .175
11	DC2-BL2 **1/7**	DC2-INETFW1 **port20**	LACP bundle member	Po11 / vPC 11 ↔ agg-inside · VLAN 3001 (second member)
12	DC2-EDGE1 **1/1**	DC2-EDGE2 **1/1**	vPC peer-link	vPC 204 peer-link Po1000 · trunk 3200
13	DC2-EDGE1 **1/2**	DC2-EDGE2 **1/2**	vPC peer-link	vPC 204 peer-link Po1000
14	DC2-EDGE1 **1/3**	DC2-EDGE2 **1/3**	vPC peer-link	vPC 204 peer-link Po1000
15	DC2-EDGE1 **1/4**	DC2-EDGE2 **1/4**	vPC peer-link	vPC 204 peer-link Po1000
16	DC2-EDGE1 **1/5**	DC2-ISP1 **Gi0-0-0-0**	LACP bundle member	Po21 / vPC 21 ↔ BE21 · VLAN 3200 OUTSIDE 10.23.14.0/24 · ISP1 .2
17	DC2-EDGE1 **1/6**	DC2-ISP2 **Gi0-0-0-0**	LACP bundle member	Po22 / vPC 22 ↔ BE21 · VLAN 3200 · ISP2 .3
18	DC2-EDGE1 **1/7**	DC2-INETFW1 **port21**	LACP bundle member	Po11 / vPC 11 ↔ agg-outside · VLAN 3200 · FW .10
19	DC2-EDGE1 **1/34**	DC2-EDGE2 **1/34**	vPC keepalive	vPC 204 keepalive · 10.23.17.6/31 · EDGE1 .6, EDGE2 .7
20	DC2-EDGE2 **1/5**	DC2-ISP1 **Gi0-0-0-1**	LACP bundle member	Po21 / vPC 21 ↔ BE21 · VLAN 3200 (second member)
21	DC2-EDGE2 **1/6**	DC2-ISP2 **Gi0-0-0-1**	LACP bundle member	Po22 / vPC 22 ↔ BE21 · VLAN 3200 (second member)
22	DC2-EDGE2 **1/7**	DC2-INETFW1 **port22**	LACP bundle member	Po11 / vPC 11 ↔ agg-outside · VLAN 3200 (second member)
23	DC2-INTFW1 **port19**	DC2-SL1 **1/7**	LACP bundle member	agg-inside ↔ Po11 / vPC 11 · VLANs 3000, 3100, 3101, 3105, 3106
24	DC2-INTFW1 **port20**	DC2-SL2 **1/7**	LACP bundle member	agg-inside ↔ Po11 / vPC 11 (second member)
25	DC2-ISP1 **Gi0-0-0-2**	DC2-ISP2 **Gi0-0-0-2**	routed	ISP cross-link 10.23.15.0/31 · ISP1 .0, ISP2 .1 · iBGP AS65050
26	DC2-L11 **1/49**	DC2-S1 **1/21**	fabric /31	10.23.4.40/31 · S1 .40, L11 .41
27	DC2-L11 **1/50**	DC2-S1 **1/22**	fabric /31	10.23.4.42/31 · S1 .42, L11 .43
28	DC2-L11 **1/51**	DC2-S2 **1/21**	fabric /31	10.23.4.132/31 · S2 .132, L11 .133
29	DC2-L11 **1/52**	DC2-S2 **1/22**	fabric /31	10.23.4.134/31 · S2 .134, L11 .135
30	DC2-S1 **1/43**	DC2-SL1 **1/1**	fabric /31	10.23.4.84/31 · S1 .84, SL1 .85
31	DC2-S1 **1/44**	DC2-SL2 **1/1**	fabric /31	10.23.4.86/31 · S1 .86, SL2 .87
32	DC2-S2 **1/43**	DC2-SL1 **1/2**	fabric /31	10.23.4.176/31 · S2 .176, SL1 .177
33	DC2-S2 **1/44**	DC2-SL2 **1/2**	fabric /31	10.23.4.178/31 · S2 .178, SL2 .179
34	DC2-SL1 **1/3**	DC2-SL2 **1/3**	vPC peer-link	vPC 203 peer-link Po1000 · trunk 2501-2502,2522,2531,2599,3000,3100-3101,3105-3106,3900
35	DC2-SL1 **1/4**	DC2-SL2 **1/4**	vPC peer-link	vPC 203 peer-link Po1000
36	DC2-SL1 **1/5**	DC2-SL2 **1/5**	vPC peer-link	vPC 203 peer-link Po1000
37	DC2-SL1 **1/6**	DC2-SL2 **1/6**	vPC peer-link	vPC 203 peer-link Po1000
38	DC2-SL1 **1/34**	DC2-SL2 **1/34**	vPC keepalive	vPC 203 keepalive · 10.23.17.4/31 · SL1 .4, SL2 .5
39	DC2-ISP1 **Gi0-0-0-3**	DC2-ISP-A **isp**	routed	192.0.2.8/30 · ISP1 .10, carrier A .9 · eBGP 65050–64496
40	DC2-ISP-A **host**	DC2-HOST-ISP-A **eth1**	host	198.18.1.0/24 · carrier A .1, host .10
41	DC2-ISP2 **Gi0-0-0-3**	DC2-ISP-B **isp**	routed	192.0.2.12/30 · ISP2 .14, carrier B .13 · eBGP 65050–64497
42	DC2-ISP-B **host**	DC2-HOST-ISP-B **eth1**	host	198.18.2.0/24 · carrier B .1, host .10

The 17 devices

Lab node	Stands in for (production)	Production model	Lab image / size	BGP AS	loopback0 / loopback1 (VTEP)	mgmt (vamd-dc2-mgmt)
DC2-S1	Spine DC2-S1	N9K-C9364D-GX2A	vrnetlab/cisco_n9kv:10.6.4.M-lite · 8 GB, 3 vCPU	65200	10.23.0.1	10.250.16.2
DC2-S2	Spine DC2-S2	N9K-C9364D-GX2A	vrnetlab/cisco_n9kv:10.6.4.M-lite · 8 GB, 3 vCPU	65200	10.23.0.2	10.250.16.3
DC2-L11	Server leaf DC2-L11 (single-homed)	Nexus 9000	vrnetlab/cisco_n9kv:10.6.4.M-lite · 8 GB, 3 vCPU	65200	10.23.0.13 / 10.23.1.11	10.250.16.14
DC2-BL1	Border leaf DC2-BL1	Nexus 9000	vrnetlab/cisco_n9kv:10.6.4.M-lite · 8 GB, 3 vCPU	65200	10.23.0.23 / 10.23.1.21, sec 10.23.2.2	10.250.16.24
DC2-BL2	Border leaf DC2-BL2	Nexus 9000	vrnetlab/cisco_n9kv:10.6.4.M-lite · 8 GB, 3 vCPU	65200	10.23.0.24 / 10.23.1.22, sec 10.23.2.2	10.250.16.25
DC2-SL1	Service leaf DC2-SL1	Nexus 9000	vrnetlab/cisco_n9kv:10.6.4.M-lite · 8 GB, 3 vCPU	65200	10.23.0.25 / 10.23.1.23, sec 10.23.2.3	10.250.16.26
DC2-SL2	Service leaf DC2-SL2	Nexus 9000	vrnetlab/cisco_n9kv:10.6.4.M-lite · 8 GB, 3 vCPU	65200	10.23.0.26 / 10.23.1.24, sec 10.23.2.3	10.250.16.27
DC2-EDGE1	Edge switch DC2-EDGE1	Nexus 9000	vrnetlab/cisco_n9kv:10.6.4.M-lite · 8 GB, 3 vCPU	none (L2)	-	10.250.16.30
DC2-EDGE2	Edge switch DC2-EDGE2	Nexus 9000	vrnetlab/cisco_n9kv:10.6.4.M-lite · 8 GB, 3 vCPU	none (L2)	-	10.250.16.31
DC2-INTFW1	Internal FW DC2-INTFW1	FortiGate	quay.io/frrouting/frr:10.6.0	65210	router-id 10.23.16.3	10.250.16.40
DC2-INETFW1	Internet FW DC2-INETFW1	FortiGate	quay.io/frrouting/frr:10.6.0	65240	router-id 10.23.16.11	10.250.16.38
DC2-ISP1	ISP router DC2-ISP1	Cisco 8201, IOS XR	quay.io/frrouting/frr:10.6.0	65050	10.23.15.17	10.250.16.36
DC2-ISP2	ISP router DC2-ISP2	Cisco 8201, IOS XR	quay.io/frrouting/frr:10.6.0	65050	10.23.15.18	10.250.16.37
DC2-ISP-A	Carrier A (not a DC2 device)	-	quay.io/frrouting/frr:10.6.0	64496	router-id 192.0.2.9	10.250.16.240
DC2-ISP-B	Carrier B (not a DC2 device)	-	quay.io/frrouting/frr:10.6.0	64497	router-id 192.0.2.13	10.250.16.241
DC2-HOST-ISP-A	A host on the internet	-	alpine:3.22	-	-	10.250.16.246
DC2-HOST-ISP-B	A host on the internet	-	alpine:3.22	-	-	10.250.16.247

Spine–leaf numbering rule: the spine takes the lower address of each /31 (pool DC2-FABRIC-P2P, 10.23.4.0/22). vPC backup underlay SVI over the peer-link: VLAN 3900, BL1 10.23.3.8 / BL2 10.23.3.9, SL1 10.23.3.10 / SL2 10.23.3.11 (/31, OSPF). The secondary loopback1 address (10.23.2.2 BL pair, 10.23.2.3 SL pair) is the vPC pair's shared anycast VTEP.

BGP sessions in the lab

Session	Who (address)	AS ↔ AS	Where / notes
iBGP EVPN (l2vpn evpn)	DC2-L11, BL1, BL2, SL1, SL2 → S1 10.23.0.1 and S2 10.23.0.2	65200 ↔ 65200	loopback0 to loopback0, spines are route reflectors; BFD multihop 1000/1000/3 configured (not up on the virtual N9Kv)
eBGP VRF TRANSIT	SL1 .1 / SL2 .2 ↔ INTFW1 .3	65200 ↔ 65210	VLAN 3000, 10.23.16.0/29; FW vrf1
eBGP VRF PROD	SL1 .1 / SL2 .2 ↔ INTFW1 .3	65200 ↔ 65210	VLAN 3100, 10.23.8.0/29; FW vrf10
eBGP VRF DEV	SL1 .9 / SL2 .10 ↔ INTFW1 .11	65200 ↔ 65210	VLAN 3101, 10.23.8.8/29; FW vrf11
eBGP VRF DC2-PROD	SL1 .17 / SL2 .18 ↔ INTFW1 .19	65200 ↔ 65210	VLAN 3102, 10.23.8.16/29; FW vrf12
eBGP VRF DC2-DEV	SL1 .25 / SL2 .26 ↔ INTFW1 .27	65200 ↔ 65210	VLAN 3103, 10.23.8.24/29; FW vrf13
eBGP VRF CITRIX	SL1 .41 / SL2 .42 ↔ INTFW1 .43	65200 ↔ 65210	VLAN 3105, 10.23.8.40/29; FW vrf15
eBGP VRF PARTNER	SL1 .49 / SL2 .50 ↔ INTFW1 .51	65200 ↔ 65210	VLAN 3106, 10.23.8.48/29; FW vrf16
eBGP VRF TRANSIT	BL1 .9 / BL2 .10 ↔ INETFW1 .11	65200 ↔ 65240	VLAN 3001, 10.23.16.8/29; FW vrf1
eBGP OUTSIDE	INETFW1 .10 ↔ ISP1 .2 / ISP2 .3	65240 ↔ 65050	VLAN 3200 through EDGE1/EDGE2, 10.23.14.0/24
iBGP ISP pair	ISP1 10.23.15.0 ↔ ISP2 10.23.15.1	65050 ↔ 65050	cross-link Gi0-0-0-2, next-hop-self
eBGP carrier A	ISP1 192.0.2.10 ↔ carrier A 192.0.2.9	65050 ↔ 64496	ISP1 sends only 203.0.113.0/24
eBGP carrier B	ISP2 192.0.2.14 ↔ carrier B 192.0.2.13	65050 ↔ 64497	ISP2 sends only 203.0.113.0/24

Underlay: OSPF process UNDERLAY, area 0.0.0.0, point-to-point on every fabric /31, loopbacks advertised. BGP password on every session: `cisco123` (a public lab constant in the templates). The Internet FW announces 203.0.113.0/24 and sends 0/0 into TRANSIT; the Internal FW leaks TRANSIT 0/0 into each tenant VRF (route-maps RM-LEAK-TRANSIT-TO-PROD/-DEV/-CITRIX/-PARTNER) and the tenant summaries into TRANSIT (RM-LEAK-TO-TRANSIT).

VRFs (tenants)

VRF	L3 VNI	L3VNI VLAN	On (lab)	FW leg (VLAN → FW VRF)	Notes
PROD (multi-site)	50001	2501	L11, SL1, SL2	3100 → INTFW1 vrf10	L11: VLANs 1010–1017 (10.16.10–17.1/24, same subnets as DC1); no host in the lab
DEV (multi-site)	50002	2502	L11, SL1, SL2	3101 → INTFW1 vrf11	L11: VLANs 1110–1118 (10.16.74–82.1/24, same subnets as DC1)
DC2-PROD	50011	2511	L11, SL1, SL2	3102 → INTFW1 vrf12	L11: VLANs 1240–1247 (10.20.40–47.1/24)
DC2-DEV	50012	2512	L11, SL1, SL2	3103 → INTFW1 vrf13	L11: VLANs 1310–1317 (10.20.74–81.1/24)
CITRIX	50022	2522	L11, SL1, SL2	3105 → INTFW1 vrf15	L11: VLANs 1500–1501 (10.20.144–145.1/24)
PARTNER	50031	2531	SL1, SL2, BL1, BL2	3106 → INTFW1 vrf16	BL: VLAN 3300 to the VPN FWs (10.23.9.0/29, AS65230), not in the lab
TRANSIT	50100	2599	SL1, SL2, BL1, BL2	3000 → INTFW1 vrf1; 3001 → INETFW1	BL: VLANs 3002–3004 to SD-WAN FW and cloud routers, not in the lab

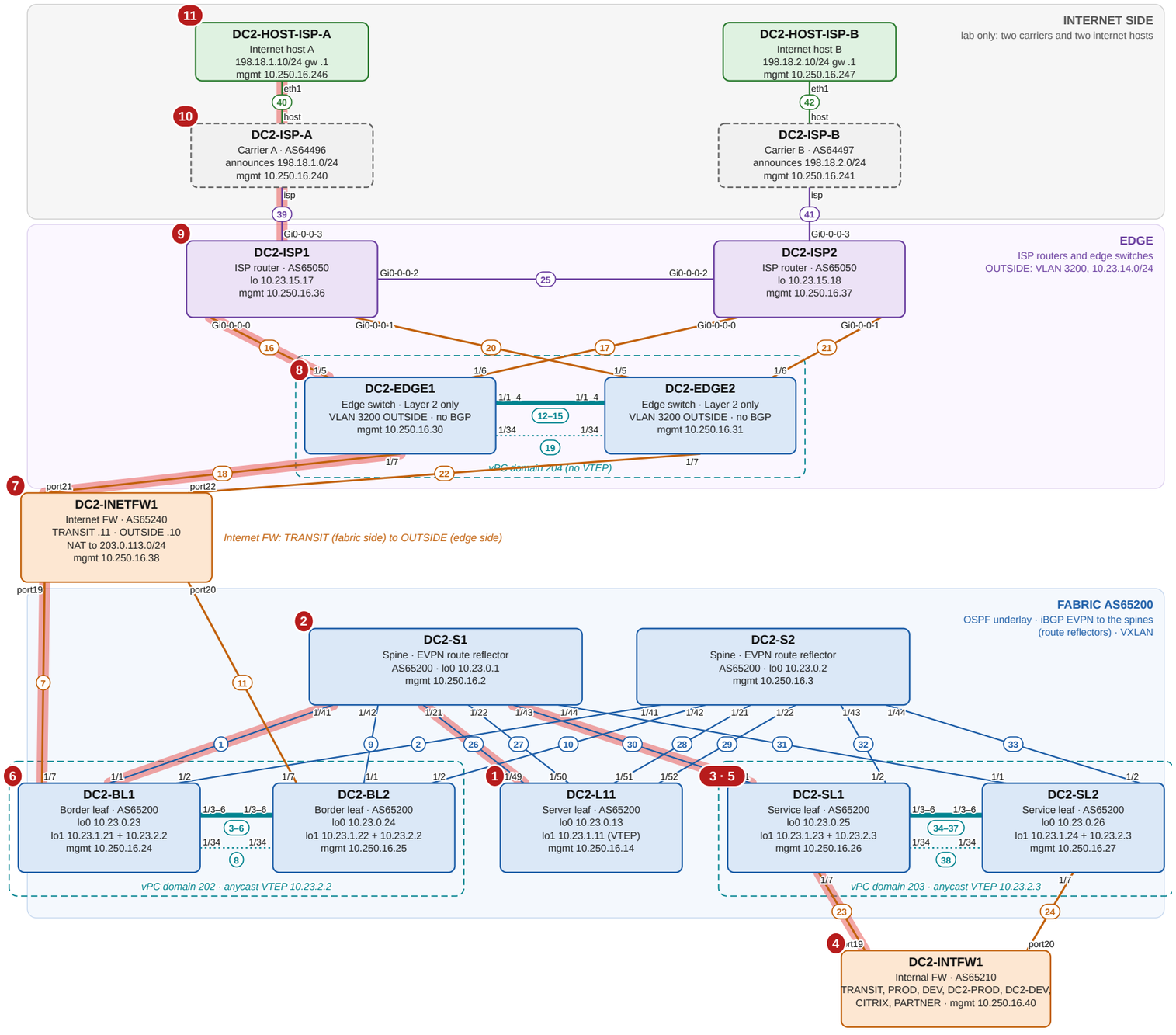
What a healthy lab should show

`lab/postcheck.py DC2` runs the checks in `lab-expected.DC2.json`. DC2 has never been run, so there is no reference result. Expect the BFD checks to fail (the virtual N9Kv does not run BFD), as in DC1. There is no PROD host, so there are no end-to-end pings.

Firewall stand-ins: what they mirror and what they do not

The FRR stand-ins copy the rendered FortiOS / IOS XR routing (interfaces, VLANs, VRFs, BGP neighbours, prefix-lists, route-maps, statics). They do not run firewall policy, IPsec, the mgmt VRF or VRRP. The production FortiOS and IOS XR configs are rendered and checked, never run.

DC2 lab path: packet walk as designed (DC2 has no PROD test host)



Red: the cables a PROD packet to the internet crosses, as designed. Every address and AS number is from the DC2 configs; nothing here was seen running, because DC2 has never been deployed.

1. **A DC2-PROD server on DC2-L11 (VLAN 1240, 10.20.40.0/24) sends to its gateway 10.20.40.1.**
   DC2-L11 has no host port in the lab; production servers attach to leaf ports. The example uses VRF DC2-PROD, DC2's own tenant (the lab Internet FW NATs 10.20.0.0/18). Anycast gateway MAC 0200.0000.00aa, as in DC1.

2. **DC2-L11 looks up the destination in VRF DC2-PROD: 0/0 learned through EVPN type-5 from the DC2 service leaves.**
   Check: `show ip route 0.0.0.0/0 vrf DC2-PROD` on DC2-L11.

3. **VXLAN with L3 VNI 50011 from 10.23.1.11 to SL1's own VTEP (10.23.1.23, advertise-pip); the spines route the outer packet over OSPF.**
   Check: `show nve peers` on DC2-L11.

4. **SL1 removes VXLAN and sends to the Internal FW 10.23.8.19 on VLAN 3102 (port-channel 11).**
   Check: `show bgp vrf DC2-PROD ipv4 unicast summary` on SL1: 10.23.8.19, AS 65210.

5. **DC2-INTFW1 moves the packet from vrf12 (DC2-PROD) to vrf1 (TRANSIT) and back to the SL pair on VLAN 3000.**
   `docker exec -it clab-vamd-dc2-p3-DC2-INTFW1 vtysh -c 'show ip route vrf vrf12'`

6. **SL1 routes in TRANSIT: 0/0 from the border leaves through EVPN, L3 VNI 50100.**
   Check: `show ip route 0.0.0.0/0 vrf TRANSIT` on SL1.

7. **BL1 removes VXLAN and sends to the Internet FW 10.23.16.11 on VLAN 3001.**
   Check: `show bgp vrf TRANSIT ipv4 unicast summary` on BL1: 10.23.16.11, AS 65240.

8. **DC2-INETFW1 source-NATs to an address of 203.0.113.0/25 and sends out of OUTSIDE (10.23.14.10, VLAN 3200).**
   The ISP routers route 203.0.113.0/25 back to 10.23.14.10 (static).

9. **EDGE1 / EDGE2 switch VLAN 3200 at Layer 2 to the ISP routers.**
   Check: `show vpc brief` on EDGE1.

10. **DC2-ISP1 (10.23.14.2) sends 198.18.1.0/24 traffic to carrier A 192.0.2.9 (eBGP 65050 ↔ 64496).**
    `docker exec -it clab-vamd-dc2-p3-DC2-ISP1 vtysh -c 'show bgp ipv4 unicast summary'`

11. **Carrier A delivers to DC2-HOST-ISP-A (198.18.1.10).**
    Test it from a host you add yourself; the DC2 lab path has no PROD host to ping from.

DC2 lab path: packet walk, the headers on every cable

Outer Ethernet + IP	UDP 4789 + VXLAN	Inner Ethernet	Inner IP	payload
VTEP → VTEP (underlay)	VNI = which VRF / VLAN	router MAC → router MAC (routed)	host → internet host	

A packet between two VTEPs (L11 ↔ SL1, SL1 ↔ BL1). On every other cable only the inner part exists (with a VLAN tag on FW and edge links).

Router MACs

Not captured: DC2 has never run. A virtual Nexus gets new MACs on every deploy; read them with `show nve peers`. The shared ones should be 0200.0a17.0202 (BL pair, 10.23.2.2) and 0200.0a17.0203 (SL pair, 10.23.2.3), by the rule seen in DC1 (0200.0a13.0202 for 10.19.2.2); not seen yet.

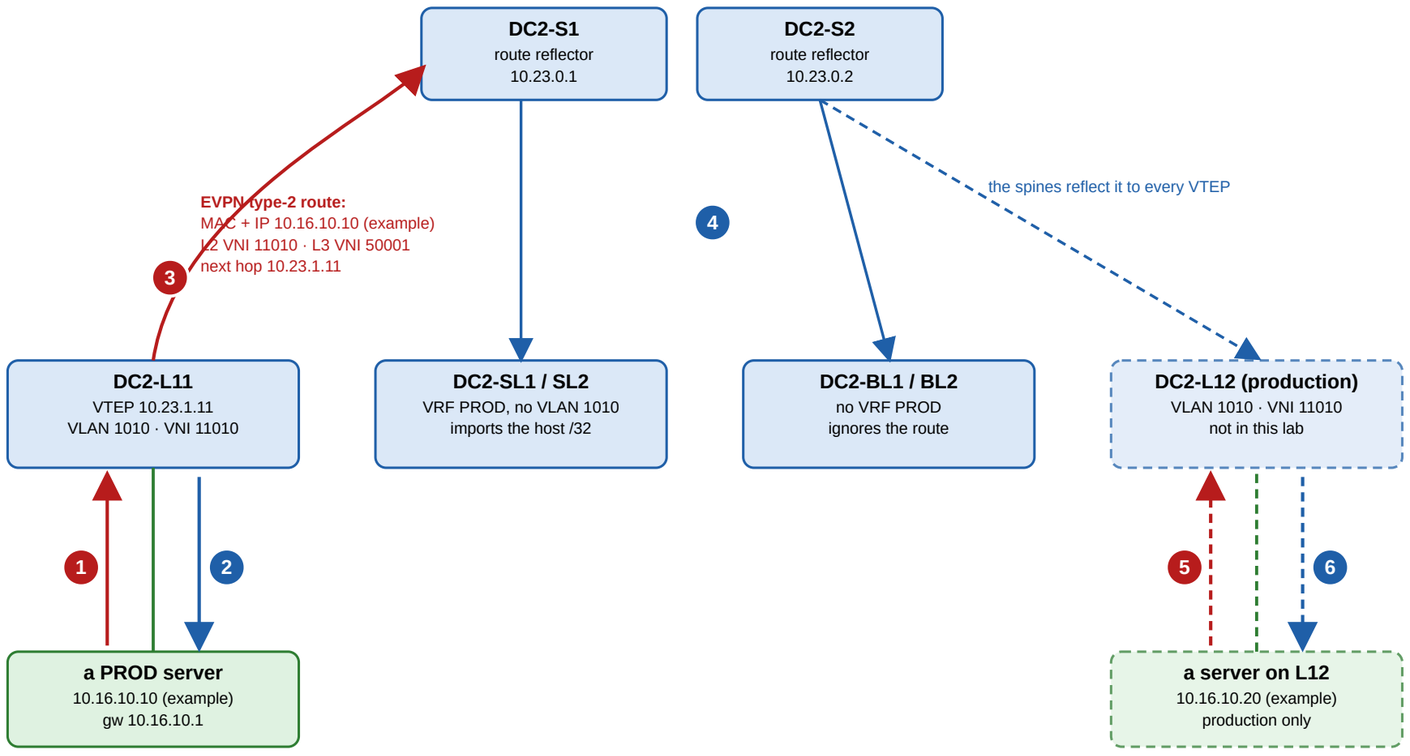
Cable (page 1 number)	802.1Q VLAN	Outer IP (VTEP → VTEP)	UDP / VXLAN	Inner Ethernet (src → dst MAC)	Inner IP (src → dst)	What the sending device did
DC2-PROD server → L11 a server port	none (access VLAN 1240 on L11)	no VXLAN on this cable		server MAC → 0200.0000.00aa (anycast gateway MAC)	DC2-PROD server → 198.18.1.10	The host sends to its gateway's MAC.
L11 → S1 cable 26 (or 27–29, ECMP)	none (routed /31)	10.23.1.11 (L11 VTEP) → 10.23.1.23 (SL1 VTEP)	UDP 4789 VXLAN VNI **50011 (L3 VNI DC2-PROD)**	L11 router MAC → SL1 router MAC	DC2-PROD server → 198.18.1.10	L11 routes in VRF DC2-PROD: 0/0 via 10.23.1.23, VNI 50011. The inner MACs become router MACs; a VXLAN header is added.
S1 → SL1 cable 30	none (routed /31)	10.23.1.11 → 10.23.1.23	UDP 4789 VXLAN VNI **50011**	L11 router MAC → SL1 router MAC	DC2-PROD server → 198.18.1.10	The spine routes only the outer IP (OSPF). Outer MACs change per hop; the inner packet is untouched.
SL1 → INTFW1 cable 23	3102 (DC2-PROD-INTFW)	no VXLAN on this cable		SL1 VLAN 3102 MAC → INTFW1 DC2-PROD-3102 MAC	DC2-PROD server → 198.18.1.10	SL1 removes VXLAN, routes in VRF DC2-PROD: 0/0 via 10.23.8.19 (eBGP AS65210).
INTFW1 → SL1 cable 23	3000 (TRANSIT-INTFW)	no VXLAN on this cable		INTFW1 TRANSIT-3000 MAC → SL VLAN 3000 MAC	DC2-PROD server → 198.18.1.10	The FW moves the packet from vrf12 (DC2-PROD) to vrf1 (TRANSIT) and sends it back to the SL pair.
SL1 → S1 cable 30	none (routed /31)	10.23.1.23 (SL1 VTEP) → 10.23.1.21 (BL1 VTEP)	UDP 4789 VXLAN VNI **50100 (L3 VNI TRANSIT)**	SL1 router MAC → BL1 router MAC	DC2-PROD server → 198.18.1.10	SL1 routes in VRF TRANSIT: 0/0 via 10.23.1.21, VNI 50100.
S1 → BL1 cable 1	none (routed /31)	10.23.1.23 → 10.23.1.21	UDP 4789 VXLAN VNI **50100**	SL1 router MAC → BL1 router MAC	DC2-PROD server → 198.18.1.10	Spine: outer IP only.
BL1 → INETFW1 cable 7	3001 (TRANSIT-INETFW)	no VXLAN on this cable		BL1 VLAN 3001 MAC → INETFW1 TRANSIT-3001 MAC	DC2-PROD server → 198.18.1.10	BL1 removes VXLAN, routes in TRANSIT: 0/0 via 10.23.16.11 (eBGP AS65240).
INETFW1 → EDGE1 → ISP1 cable 18, then 16	3200 (OUTSIDE)	no VXLAN on this cable		INETFW1 OUTSIDE-3200 MAC → ISP1 BE21.3200 MAC	203.0.113.x (NAT) → 198.18.1.10	Source NAT: the private address becomes one address of 203.0.113.1–126. EDGE1 switches VLAN 3200 at Layer 2 only.
ISP1 → carrier A cable 39	none (routed /30)	no VXLAN on this cable		ISP1 Gi0-0-0-3 MAC → carrier A isp port MAC	203.0.113.x → 198.18.1.10	ISP1 routes 198.18.1.0/24 learned from carrier A (eBGP 65050 ↔ 64496).
carrier A → HOST-ISP-A cable 40	none	no VXLAN on this cable		carrier A host port MAC → host A eth1 MAC	203.0.113.x → 198.18.1.10	Delivered.

Where the values come from. VTEP addresses, VNIs, VLANs, the anycast MAC and the next hops: the DC2 configs rendered by Infrahub (designed values; DC2 has never run). MACs in plain words were not captured.

Why VTEP 10.23.1.23 and not the SL pair's 10.23.2.3. The pairs run `advertise-pip`: routes a vPC switch learns itself (here the FW's 0/0) are sent with the switch's own VTEP address and router MAC, so traffic goes to the switch that has the route. Host MACs behind a vPC pair use the shared address 10.23.2.3 / 10.23.2.2 with the virtual router MAC.

Return. Carrier A → ISP1, which routes 203.0.113.0/25 to the Internet FW 10.23.14.10 (static) → EDGE (VLAN 3200) → INETFW1 reverses the NAT → BL (TRANSIT; tenant summary learned from the SL pair through EVPN) → SL → INTFW1 → SL (VRF DC2-PROD; the server's host route is an EVPN type-2 from L11) → L11 → host.

DC2 lab path: the first ARP and how the fabric learns a host



1 ARP request (broadcast): who has 10.16.10.1? **2** L11 answers itself: 10.16.10.1 is at 0200.0000.00aa (anycast gateway, on every leaf)

5 the L12 server asks for 10.16.10.10 (example) **6** L12 answers from its EVPN table with the host's MAC (ARP suppression): the request never crosses the fabric

DC2 lab path: no PROD host and one server leaf, so none of these steps can be seen in the DC2 lab path as it is. Steps 1–4 work as soon as a host is attached to a DC2-L11 port in VLAN 1010 (the same change as DC1's Eth1/1, P3-39). The commands to look at each step are on DC1's page 6.

Config lines behind it (every leaf)

```
fabric forwarding anycast-gateway-mac 0200.0000.00aa
interface Vlan1010
  fabric forwarding mode anycast-gateway
interface nve1
  global suppress-arp
  global ingress-replication protocol bgp
  member vni 11010
```

ARP suppression needs the TCAM region the `carve` step sets on the virtual Nexus (`arp-ether 256 double-wide`).

Dashed boxes and arrows: production behaviour that this lab cannot show. Option 5 (entire DC2) has all 20 leaves; it needs a second test host to show steps 5–6.

DC2 lab path: lab cables vs production cables, and what is configured but missing

Each lab cable is a real DC2 cable (from the vendor cable schedule rendered by Infrahub)

#	Lab cable (containerlab names)	Production cable label	Media	Length	Route (patch panel ports)
1	BL1 **eth1** ↔ S1 **eth41**	DC2-S1:Eth1/41-DC2-BL1:Eth1/1	dac	2 m	same rack DC2-B04
2	BL1 **eth2** ↔ S2 **eth41**	DC2-S2:Eth1/41-DC2-BL1:Eth1/2	smf	26 m	DC2-B08-PP42:83 → DC2-B04-PP42:83
3	BL1 **eth3** ↔ BL2 **eth3**	DC2-BL1:Eth1/3-DC2-BL2:Eth1/3	smf	26 m	DC2-B04-PP42:1 → DC2-B08-PP42:1
4	BL1 **eth4** ↔ BL2 **eth4**	DC2-BL1:Eth1/4-DC2-BL2:Eth1/4	smf	26 m	PP42:2 → PP42:2
5	BL1 **eth5** ↔ BL2 **eth5**	DC2-BL1:Eth1/5-DC2-BL2:Eth1/5	smf	26 m	PP42:3 → PP42:3
6	BL1 **eth6** ↔ BL2 **eth6**	DC2-BL1:Eth1/6-DC2-BL2:Eth1/6	smf	26 m	PP42:4 → PP42:4
7	BL1 **eth7** ↔ INETFW1 **port19**	DC2-BL1:Eth1/7-DC2-INETFW1:port19	dac	3 m	same rack DC2-B04
8	BL1 **eth34** ↔ BL2 **eth34**	DC2-BL1:Eth1/34-DC2-BL2:Eth1/34	smf	26 m	PP42:9 → PP42:9
9	BL2 **eth1** ↔ S1 **eth42**	DC2-S1:Eth1/42-DC2-BL2:Eth1/1	smf	26 m	DC2-B04-PP42:80 → DC2-B08-PP42:40
10	BL2 **eth2** ↔ S2 **eth42**	DC2-S2:Eth1/42-DC2-BL2:Eth1/2	dac	2 m	same rack DC2-B08
11	BL2 **eth7** ↔ INETFW1 **port20**	DC2-BL2:Eth1/7-DC2-INETFW1:port20	smf	27 m	DC2-B08-PP42:10 → DC2-B04-PP42:10
12	EDGE1 **eth1** ↔ EDGE2 **eth1**	DC2-EDGE1:Eth1/1-DC2-EDGE2:Eth1/1	smf	26 m	PP42:22 → PP42:22
13	EDGE1 **eth2** ↔ EDGE2 **eth2**	DC2-EDGE1:Eth1/2-DC2-EDGE2:Eth1/2	smf	26 m	PP42:23 → PP42:23
14	EDGE1 **eth3** ↔ EDGE2 **eth3**	DC2-EDGE1:Eth1/3-DC2-EDGE2:Eth1/3	smf	26 m	PP42:24 → PP42:24
15	EDGE1 **eth4** ↔ EDGE2 **eth4**	DC2-EDGE1:Eth1/4-DC2-EDGE2:Eth1/4	smf	26 m	PP42:25 → PP42:25
16	EDGE1 **eth5** ↔ ISP1 **Gi0-0-0-0**	DC2-EDGE1:Eth1/5-DC2-ISP1:FH0/0/0/0	dac	2 m	same rack DC2-B04
17	EDGE1 **eth6** ↔ ISP2 **Gi0-0-0-0**	DC2-EDGE1:Eth1/6-DC2-ISP2:FH0/0/0/0	smf	26 m	DC2-B04-PP42:26 → DC2-B08-PP42:26
18	EDGE1 **eth7** ↔ INETFW1 **port21**	DC2-EDGE1:Eth1/7-DC2-INETFW1:port21	dac	3 m	same rack DC2-B04
19	EDGE1 **eth34** ↔ EDGE2 **eth34**	DC2-EDGE1:Eth1/34-DC2-EDGE2:Eth1/34	smf	26 m	PP42:30 → PP42:30
20	EDGE2 **eth5** ↔ ISP1 **Gi0-0-0-1**	DC2-EDGE2:Eth1/5-DC2-ISP1:FH0/0/0/2	smf	26 m	DC2-B08-PP42:31 → DC2-B04-PP42:31
21	EDGE2 **eth6** ↔ ISP2 **Gi0-0-0-1**	DC2-EDGE2:Eth1/6-DC2-ISP2:FH0/0/0/2	dac	2 m	same rack DC2-B08
22	EDGE2 **eth7** ↔ INETFW1 **port22**	DC2-EDGE2:Eth1/7-DC2-INETFW1:port22	smf	27 m	DC2-B08-PP42:32 → DC2-B04-PP42:32
23	INTFW1 **port19** ↔ SL1 **eth7**	DC2-SL1:Eth1/7-DC2-INTFW1:port19	dac	3 m	same rack DC2-B04
24	INTFW1 **port20** ↔ SL2 **eth7**	DC2-SL2:Eth1/7-DC2-INTFW1:port20	smf	27 m	DC2-B08-PP42:94 → DC2-B04-PP42:94
25	ISP1 **Gi0-0-0-2** ↔ ISP2 **Gi0-0-0-2**	DC2-ISP1:FH0/0/0/4-DC2-ISP2:FH0/0/0/4	smf	26 m	DC2-B04-PP42:39 → DC2-B08-PP42:39
26	L11 **eth49** ↔ S1 **eth21**	DC2-S1:Eth1/21-DC2-L11:Eth1/49	smf	28 m	DC2-B04-PP42:60 → DC2-C01-PP42:1
27	L11 **eth50** ↔ S1 **eth22**	DC2-S1:Eth1/22-DC2-L11:Eth1/50	smf	28 m	DC2-B04-PP42:61 → DC2-C01-PP42:2
28	L11 **eth51** ↔ S2 **eth21**	DC2-S2:Eth1/21-DC2-L11:Eth1/51	smf	31 m	DC2-B08-PP42:63 → DC2-C01-PP42:3
29	L11 **eth52** ↔ S2 **eth22**	DC2-S2:Eth1/22-DC2-L11:Eth1/52	smf	31 m	DC2-B08-PP42:64 → DC2-C01-PP42:4
30	S1 **eth43** ↔ SL1 **eth1**	DC2-S1:Eth1/43-DC2-SL1:Eth1/1	dac	2 m	same rack DC2-B04
31	S1 **eth44** ↔ SL2 **eth1**	DC2-S1:Eth1/44-DC2-SL2:Eth1/1	smf	26 m	DC2-B04-PP42:81 → DC2-B08-PP42:41
32	S2 **eth43** ↔ SL1 **eth2**	DC2-S2:Eth1/43-DC2-SL1:Eth1/2	smf	26 m	DC2-B08-PP42:84 → DC2-B04-PP42:84
33	S2 **eth44** ↔ SL2 **eth2**	DC2-S2:Eth1/44-DC2-SL2:Eth1/2	dac	2 m	same rack DC2-B08
34	SL1 **eth3** ↔ SL2 **eth3**	DC2-SL1:Eth1/3-DC2-SL2:Eth1/3	smf	26 m	PP42:88 → PP42:88
35	SL1 **eth4** ↔ SL2 **eth4**	DC2-SL1:Eth1/4-DC2-SL2:Eth1/4	smf	26 m	PP42:89 → PP42:89
36	SL1 **eth5** ↔ SL2 **eth5**	DC2-SL1:Eth1/5-DC2-SL2:Eth1/5	smf	26 m	PP42:90 → PP42:90
37	SL1 **eth6** ↔ SL2 **eth6**	DC2-SL1:Eth1/6-DC2-SL2:Eth1/6	smf	26 m	PP42:91 → PP42:91
38	SL1 **eth34** ↔ SL2 **eth34**	DC2-SL1:Eth1/34-DC2-SL2:Eth1/34	smf	26 m	PP42:93 → PP42:93
39	ISP1 **Gi0-0-0-3** ↔ ISP-A **isp**	DC2-ISP1:FH0/0/0/6 carrier handoff (ISP-A)	carrier	-	outside the site
40	ISP-A **host** ↔ HOST-ISP-A **eth1**	lab only	-	-	-
41	ISP2 **Gi0-0-0-3** ↔ ISP-B **isp**	DC2-ISP2:FH0/0/0/6 carrier handoff (ISP-B)	carrier	-	outside the site
42	ISP-B **host** ↔ HOST-ISP-B **eth1**	lab only	-	-	-

Port names: containerlab eth*N* = Ethernet1/*N* on the Nexus; the ISP stand-ins' Gi0-0-0-*k* = FourHundredGigE0/0/0/*2k* on the Cisco 8201 (0 → FH0/0/0/0, 1 → FH0/0/0/2, 2 → FH0/0/0/4 cross-link, 3 → FH0/0/0/6 carrier). Same-rack cables are DAC; cross-rack cables are single-mode fibre through the patch racks (length = jumpers + trunks). DC2 has no added host port.

Configured on a lab device, but the far end is not in the lab (expected to be down)

Lab device	Ports with no far end	BGP sessions that stay down
DC2-S1 / S2	Eth1/1–1/20, 1/23–1/40 (L1–L10, L12–L20), Eth1/45–46 (BGW1/2)	21 of 26 EVPN route-reflector clients (only L11, BL1/2, SL1/2 are here)
DC2-BL1 / BL2	Eth1/8 INETFW2, 1/9–1/10 SDWANFW1/2, 1/11–1/12 VPNFW1/2, 1/13–1/14 CLOUD1/2	TRANSIT: 10.23.16.19 (SD-WAN FW AS65220), 10.23.16.27 / .35 (cloud routers AS65270); PARTNER: 10.23.9.3 (VPN FW AS65230)
DC2-SL1 / SL2	Eth1/8 INTFW2 (port-channel 12)	none
DC2-EDGE1 / EDGE2	Eth1/8 INETFW2, 1/9–1/12 SDWANFW1/2 and VPNFW1/2	none (Layer 2 only)
DC2-INETFW1	port15–16 agg-dmz (to DMZ1/DMZ2, VLANs 1400–1415)	none; the DMZ subnets 10.20.128–143.0/24 exist on the FW with no host
DC2-ISP1 / ISP2	none in the lab path	VRRP not run on FRR
DC2-L11	none (single-homed leaf)	none

These are the reason `show interface status`, `show vpc brief` and `show bgp … summary` show ports, vPCs and sessions down in a healthy lab. `postcheck.py` checks only the sessions in `lab-expected`, so they do not count as failures.

VRFs and their segments

VRF	L3 VNI	L3VNI VLAN	Tenant VLANs	L2 VNIs	Subnets (gateway .1)	On lab devices	FW leg
PROD (multi-site)	50001	2501	1010–1017	11010–11017	10.16.10–17.0/24	L11; SL1/2 (L3 only)	3100 → INTFW1 vrf10
DEV (multi-site)	50002	2502	1110–1118	11110–11118	10.16.74–82.0/24	L11; SL1/2 (L3 only)	3101 → INTFW1 vrf11
DC2-PROD	50011	2511	1240–1247	11240–11247	10.20.40–47.0/24	L11; SL1/2 (L3 only)	3102 → INTFW1 vrf12
DC2-DEV	50012	2512	1310–1317	11310–11317	10.20.74–81.0/24	L11; SL1/2 (L3 only)	3103 → INTFW1 vrf13
CITRIX	50022	2522	1500–1501	11500–11501	10.20.144–145.0/24	L11; SL1/2 (L3 only)	3105 → INTFW1 vrf15
PARTNER	50031	2531	–	–	–	SL1/2, BL1/2 (L3 only)	3106 → INTFW1 vrf16; BL 3300 → VPN FW (not in lab)
TRANSIT	50100	2599	–	–	–	SL1/2, BL1/2 (L3 only)	3000 → INTFW1 vrf1; BL 3001 → INETFW1
REPLICATION	50041	2541	1599	11599	10.20.176.0/24	not on any lab device	–

Rule seen in the data: L2 VNI = 10000 + VLAN. Every tenant subnet's gateway is the anycast address .1 with MAC 0200.0000.00aa on every leaf that has the VLAN. PROD and DEV are multi-site VRFs: DC2 uses the same VLANs and subnets as DC1 for them. L3-only means the VRF exists with its L3 VNI but no tenant VLAN (the switch routes for it, hosts are elsewhere).

VTEPs in the lab

VTEP	Own address (loopback1)	Shared vPC address (secondary)	Router MAC	VNIs it carries
DC2-L11	10.23.1.11	–	new on every deploy	PROD, DEV, DC2-PROD, DC2-DEV, CITRIX + 35 L2 VNIs
DC2-SL1	10.23.1.23	10.23.2.3	new on every deploy	the 5 above + PARTNER, TRANSIT
DC2-SL2	10.23.1.24	10.23.2.3	new on every deploy	same as SL1
DC2-BL1	10.23.1.21	10.23.2.2	new on every deploy	PARTNER, TRANSIT
DC2-BL2	10.23.1.22	10.23.2.2	new on every deploy	same as BL1

Shared addresses should carry the virtual router MACs 0200.0a17.0203 (SL pair) and 0200.0a17.0202 (BL pair), by the DC1 rule; not seen yet. BUM traffic: ingress replication over BGP (EVPN type-3), no multicast in the underlay. Check: `show nve vni`, `show nve peers`, `show bgp l2vpn evpn summary` (type-2 / type-5 counts).

What you need

Item	Value
Memory	About 80 GB free for the lab: 9 Nexus × 8 GB (QEMU_MEMORY 8192) = 72 GB plus 8 FRR / Alpine containers. Never run; the DC1 lab of the same size ran on AWS m8i.8xlarge (32 vCPU, 123 GB).
CPU	Each Nexus 3 vCPU (QEMU_SMP 3); nested virtualization (KVM) on the host.
Software	containerlab 0.79, Docker, image `vrnetlab/cisco_n9kv:10.6.4.M-lite` (built from your own Cisco download), `quay.io/frrouting/frr:10.6.0`, `alpine:3.22`.
Infrahub	Running somewhere you can reach for `fetch` (laptop is fine); the lab host does not need it after fetch.
Time	Expect about 27 minutes from `deploy` to a fully configured lab (DC1's time).

Bring it up

```
# files from Infrahub, compared with the repo
.venv/bin/python lab/labctl.py fetch DC2 --compare-baselines
# topology, files, images, memory
.venv/bin/python lab/labctl.py validate DC2
# boots all 17 nodes (minimal Nexus config)
.venv/bin/python lab/labctl.py deploy DC2
# full rendered configs over NX-API
.venv/bin/python lab/labctl.py push DC2
# TCAM for ARP suppression, save, reload, re-push
.venv/bin/python lab/labctl.py carve DC2
# checks from lab-expected.DC2.json (page 3)
.venv-lab/bin/python lab/postcheck.py DC2
```

Take it down

```
.venv/bin/python lab/labctl.py destroy DC2
```

Log in

Device kind	How
Nexus (9)	`ssh vamdadmin@<mgmt address>` (page 1 / page 2), password from `ztp/make-secret.sh`
FRR stand-ins (6)	`docker exec -it clab-vamd-dc2-p3-DC2-INTFW1 vtysh` (any FRR node name)
Hosts (2)	`docker exec -it clab-vamd-dc2-p3-DC2-HOST-ISP-A sh`

Stop and start one device

The topology file containerlab uses is `lab/build/DC2/topology.boot.yml` (`topology.override.yml` when `deploy` ran with `--memory`, as on a laptop). Find it with `docker inspect -f '{{index .Config.Labels "clab-topo-file"}}' clab-vamd-dc2-p3-DC2-L11`.

Device kind	Stop	Start	Tested
FRR stand-ins, hosts	`containerlab stop -t <file> --node DC2-INTFW1`	`containerlab start -t <file> --node DC2-INTFW1`	**DC1 laptop test 19 Oct: back in seconds, config, VRFs, addresses and routes the same**
Nexus	`containerlab stop -t <file> --node DC2-L11` works	Per-node start did not bring the switch back (40 min). Use destroy + deploy, push, carve.	**DC1 laptop test 19 Oct: start failed**

Cut a cable without stopping anything

```
# Nexus side, logged in to the switch (example: L11 uplink to S1, cable 25)
configure terminal
interface Ethernet1/49
  shutdown            # "no shutdown" puts it back
# FRR side (example: Internal FW member to SL1, cable 22)
docker exec clab-vamd-dc2-p3-DC2-INTFW1 ip link set port19 down    # "up" puts it back
```

Smaller labs from the same files

Option	Nodes	Memory	Status
1 · one switch	DC2-L11 (`--node-filter DC2-L11 --memory 6144`)	about 7 GB free	Runs on the 12 GB laptop
2 · DC1 lab path	18	about 80 GB	Ran on AWS 8 Oct
3 · DC2 lab path (this document, 9 pages)	17	about 80 GB	Rendered, never run
4 · DC2 + DC2 lab paths	35 (two separate labs)	about 160 GB	Not run
5 · entire DC2 production	46 + stand-ins	about 290 GB	Being written